ELSEVIER

Earth and Planetary Science Letters 143 (1996) 37–48

EPSL

Magnetostratigraphy of the Eocene/Oligocene boundary in a short drill-core

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Received 16 January 1996; revised 25 June 1996; accepted 26 June 1996

Abstract

A 39.4 m long, 10 cm diameter vertical core was drilled through the Eocene/Oligocene boundary close to the Massignano quarry stratotype section near Ancona, Italy. Total recovery was about 95% and 73% of the core was intact. It was oriented by comparing the dip of bedding planes visible on the core surface with bedding measured in the nearby quarry outcrop. The core was correlated to the quarry section by linear regression of the depths of four biotite-rich layers in each section. The fissile condition of the *Scaglia cinerea* marls made sampling difficult, but 260 samples were obtained at an average separation of 12 cm in this and the underlying *Scaglia variegata* formation. Progressive alternating field and thermal demagnetization isolated two magnetization components in most samples. A stable component with unblocking temperatures of 300–540°C and coercivities over 20 mT is probably carried by magnetite. A stratigraphic plot of the ChRM directions shows well-defined normal and reversed magnetozones. The polarity sequence correlates well with chrons C12r to C16n-2. A few single-sample normal magnetozones within chron C16n.1-r have no corollary in the geomagnetic polarity timescale. The magnetozones corresponding to chrons C12r or C13r exhibit no short subchrons that might account for low-amplitude, short-wavelength magnetic anomalies reported in this part of the marine magnetic record.

Keywords: paleomagnetism; magnetostratigraphy; reversals; Eocene; Oligocene; Ancona Italy; Scaglia Formation

1. Introduction

The history of geomagnetic polarity since the Late Jurassic is defined from the analysis of lineated oceanic magnetic anomalies with generally large amplitudes and well defined shapes. Cande and Kent [1] critically reappraised the oceanic magnetic anomaly record for the Late Cretaceous and Cenozoic and produced a definitive polarity sequence. By calibrating the polarity sequence based on securely identified chrons at well dated tie-levels they developed a new polarity timescale [1]. A later, definitive version of the timescale (hereafter referred to as CK95) incorporated improved tie-level ages [2].

Magnetostratigraphic investigations of the geomagnetic polarity record have been carried out successfully in various types of marine and continental sediments and sedimentary rocks. The studies, made in land outcrops and in cores drilled from the ocean floor, have focused primarily on verifying the marine reversal record by establishing it in coeval or over-

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PII S0012-821X(96)00136-7

lapping sections. The magnetostratigraphies of pelagic limestone sequences have confirmed the main features of the polarity sequences in the Late Jurassic and Early Cretaceous [3,4] and in the Late Cretaceous and Paleogene [5,6]. Polarity timescales are formed by associating numeric ages with well dated paleontological stage boundaries that can be clearly correlated with the polarity sequence. The Eocene/Oligocene (E/O) boundary fulfils these criteria and is an important Cenozoic calibration level in recent timescales [1,2]. An interpolated age of 33.7 ± 0.5 Ma was obtained for this stage boundary from K–Ar and Rb–Sr dating of biotites from ash layers in Central Italian magnetostratigraphic sections [7,8].

Magnetostratigraphic studies that included the E/O boundary have been carried out in Central Italy at two locations (Fig. 1): in the Contessa valley near Gubbio [9,10], and in the Massignano quarry near Ancona [11–13]. In each location the boundary falls in the lower part of the *Scaglia cinerea* formation, just above its gradual transition to the underlying *Scaglia variegata* formation. The lithology of the *Scaglia cinerea* is a succession of dark marls and marly limestones; the *Scaglia variegata* consists of mainly limestone beds with variable hue. The formations are more weathered in the Massignano quarry than in the Contessa sections, where the outcrop is fresh. The *Scaglia cinerea* at Massignano has a greenish brown color, whereas in the Contessa valley its color is a dark bluish-grey.

In their revision of the Late Cretaceous–Cenozoic marine magnetic anomaly record Cande and Kent [1] described 50 events that appear on magnetic profiles as low-amplitude, short-wavelength anomalies. The interpretation of these 'tiny wiggles' is ambiguous [14]. They cannot be ascribed uniquely to short polarity chrons, even on high resolution magnetic profiles. Acknowledging their uncertain origin, Cande and Kent [15] called the events 'cryptochrons' and suggested that they are probably not polarity chrons but are caused by intensity fluctuations of the paleomagnetic field.

Lowrie and Lanci [13] evaluated critically the magnetostratigraphies of E/O boundary sections in the Contessa valley and in the Massignano quarry (Fig. 1). They concluded that there was no unambiguous evidence for short chrons in these magne-

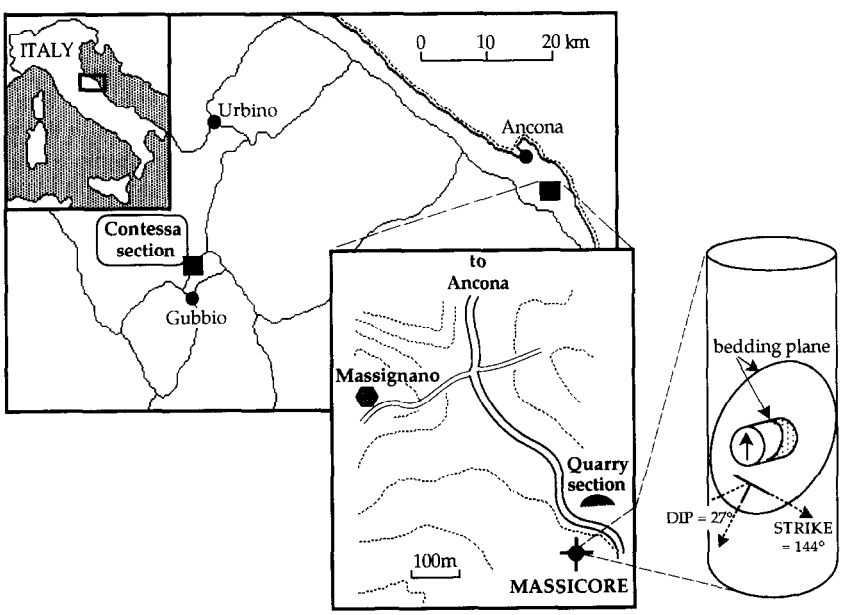


Fig. 1. Locations of Late Eocene/Early Oligocene magnetostratigraphic sections in the Massignano and Contessa quarries in north-central Italy. The MASSICORE drillhole was located 110 m south of the Massignano quarry section; bedding laminations on the core surface were used to orient paleomagnetic samples.

tostratigraphic sections. The Massignano quarry section is complete paleontologically [16]. It has been proposed and accepted as the stratotype section for the E/O boundary [17,18]. Unfortunately, the magnetic properties of the sediments are less than ideal. Due to strong weathering, closely spaced, intact samples of the *Scaglia cinerea* and *Scaglia variegata* are hard to obtain in the outcrop. In both formations weathering has also reduced the intensity of the remanent magnetization and may have added secondary components. To obviate these problems it was decided to obtain a drilled core through the formations.

2. Core description and sampling

The core (hereafter called the MASSICORE) was commissioned by Dr. Fabio Pallotta and Dr. Piero Beer of the Ordine Regionale dei Geologi delle

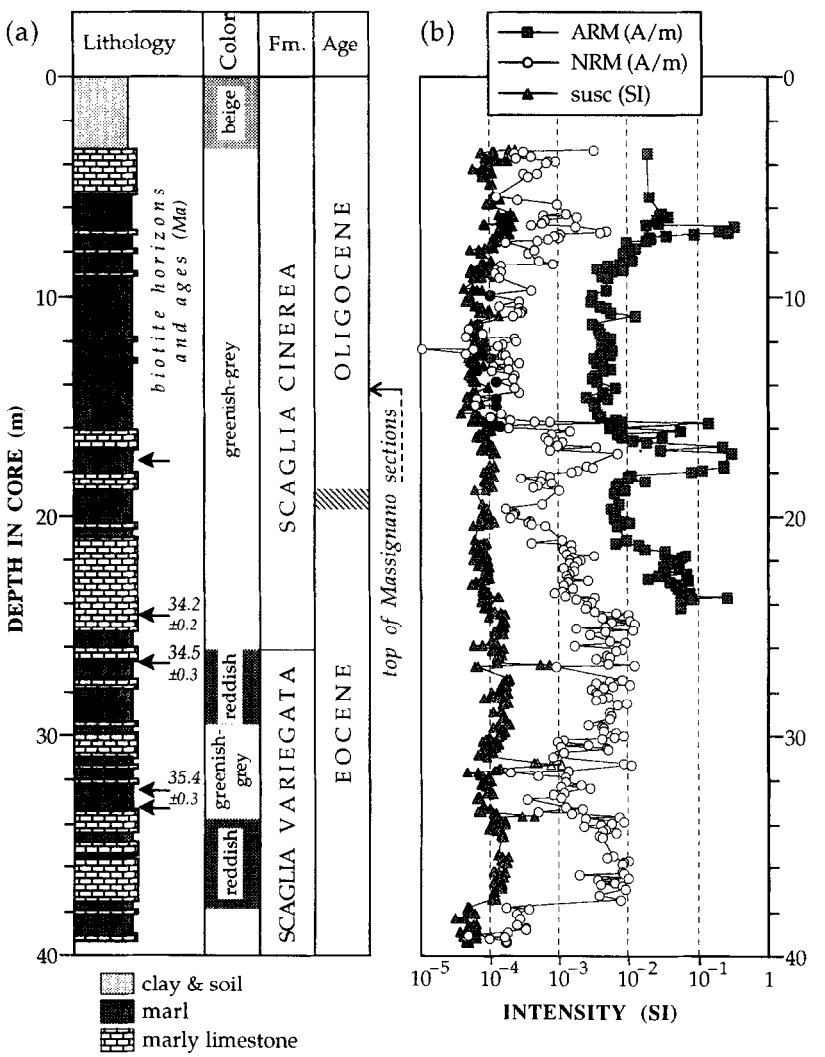


Fig. 2. (a) Lithology and preliminary dating of the MASSICORE; the position of the Eocene/Oligocene boundary was interpolated by lithostratigraphic correlation with the nearby paleontologically dated stratotype section; the numbers at biotite-rich horizons give the mean ages of these layers in the Massignano quarry sections [22]. (b) Variations of magnetic susceptibility, natural remanent magnetization (NRM) and anhysteretic remanent magnetization (ARM) with depth in the core.

Marche for the promotion of high-resolution stratigraphic and paleoecologic studies of the Massignano E/O boundary section. It is permanently stored at the Osservatorio Geologico di Coldigioco in Frontale (Italy). The 39.4 m deep, 10 cm diameter core was drilled vertically about 110 m south of the stratotype section; the site coordinates are 42°32′09.6″N, 13°35′34.3″E. The core was logged in detail [19] with respect to core recovery, lithology, color and marker beds represented by biotite-rich layers (Fig. 2). The total core recovery was 94.6% and 72.7% was recovered intact. Core sections usually broke along the surfaces of soft marl layers or clay beds. The stratigraphic interval sampled by the MASSICORE overlaps the 23 m thick Massignano stratotype section and extends it upward by about 12 m further into the *Scaglia cinerea* formation. The uppermost 3 m consist of deeply altered marls, loose clay and soils that are not useful stratigraphically. The rest of the core is fresh and suitable for magnetostratigraphic, paleontologic and sedimentologic studies.

Standard 2.5 cm diameter paleomagnetic samples were drilled perpendicular to the core axis at an average spacing of 12–15 cm, the sampling interval being dictated by the condition and continuity of the core. A total of 260 samples was taken, most of which were long enough to give two specimens. Bedding laminations were visible on the surfaces of the core and the drilled samples, enabling azimuthal orientation of core segments by comparison with the attitude of bedding in local outcrops, where it has a strike of N144°E and a dip of 20° SW. The dip of bedding in the core was checked by using the visible laminations on sample surfaces. The orthogonal distances to a lamination along and normal to the sample axis were measured with a millimeter scale to obtain the tangent of the dip angle. The calculation was made on about 20 samples, which gave a consistent bedding dip of 27° ± 1° in the core. This is very similar to the dip of 20° measured in the quarry outcrop. The small difference in dip takes place gently over a distance of about 100 m, and may indicate a locally plunging fold axis. The stratification is regular and free of faults and small-scale folding.

The precise location of the E/O boundary in the Massignano quarry was transferred to the core by lithologic correlation. Subtle differences in lithology

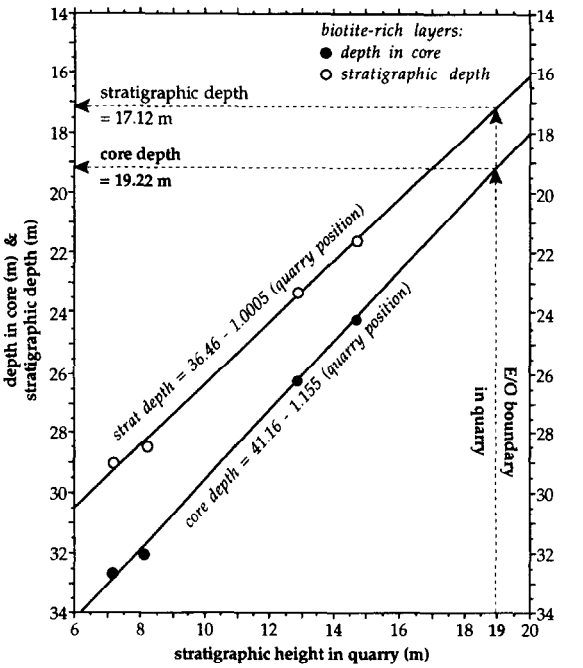


Fig. 3. Linear regression of the positions of biotite-rich layers in the Massignano quarry section with their depths in the MASSICORE and with their computed stratigraphic depths.

are more apparent in the core than in the quarry outcrop where weathering obscures details. Variations in lithology and color provided first-order matching between the core and the quarry. The positions of four recognizable biotite-rich layers in the core and the quarry outcrop permitted precise correlation of the two sections by linear regression (Fig. 3). In the Massignano quarry stratotype section the E/O boundary is located at a stratigraphic level of 19 m [16]. This corresponds to a core depth of 19.2 m in the MASSICORE, and an equivalent stratigraphic depth (normal to bedding) of 17.1 m.

The Massignano quarry section is the global stratotype section and point (GSSP) for the E/O boundary. K–Ar and Rb–Sr dating of biotite-rich layers in Central Italian magnetostratigraphic sections [8] gave an age of 33.7 Ma for the boundary, which has been incorporated as a calibration-point in the latest polarity timescales [1,2]. Additional results from $^{40}Ar/^{39}Ar$ dating of biotites [20] and U/Pb dating of zircons and monazites [21] were combined with the earlier data, yielding revised mean ages of 35.4 ± 0.3 Ma, 34.5 ± 0.3 Ma and 34.2 ± 0.2 Ma for the biotite

horizons at 7.2 m, 12.9 m and 14.7 m, respectively, in the Massignano quarry section [22]. The corresponding depths in the MASSICORE are 32.8 m, 26.3 m and 24.2 m, respectively (Fig. 2). By extrapolating the best-fit line through these points to the E/O boundary a revised age of 33.5 ± 0.3 Ma is obtained. The overlapping confidence limits of this and the earlier estimate show that they are not different statistically.

3. Magnetic properties

The ferromagnetic mineralogy of the Massignano quarry limestones has been reported previously [13]. The analysis involved successively (1) progressive acquisition of isothermal remanent magnetization (IRM) in fields up to 1.2 T, (2) imposition of a multicomponent IRM in different coercivity ranges, and (3) subsequent thermal demagnetization of the

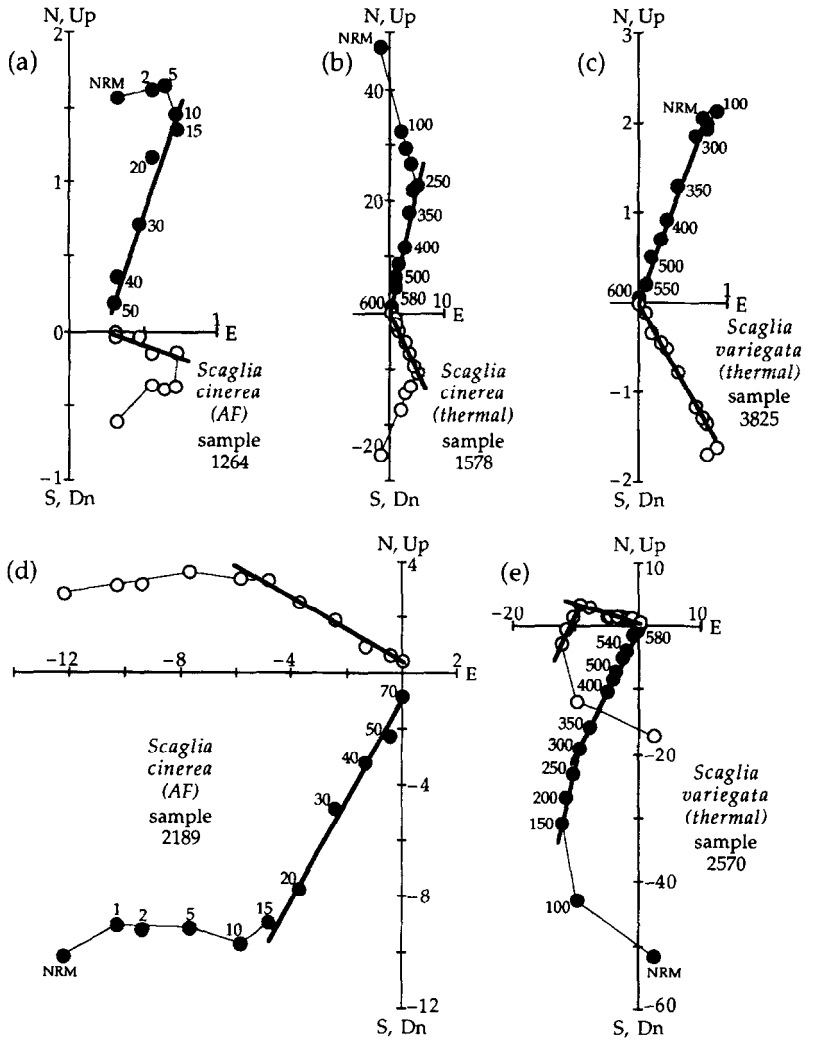


Fig. 4. Vector diagrams for (a) AF demagnetization of normal *Scaglia cinerea*, (b) thermal demagnetization of normal *Scaglia cinerea*, (c) thermal demagnetization of normal *Scaglia variegata*, (d) AF demagnetization of reversed *Scaglia cinerea*, and (e) thermal demagnetization of reversed *Scaglia variegata*. Numbers on the plots indicate peak demagnetizing field in mT or heating temperature in °C. Dots and circles represent projections on the horizontal and E–W vertical planes, respectively.

compound IRM [23]. The only significant magnetic mineral in the greenish-grey *Scaglia cinerea* is magnetite, while the multi-hued *Scaglia variegata* contains both magnetite and hematite.

The intensity of the natural remanent magnetization (NRM) is about an order of magnitude weaker in the *Scaglia cinerea* (mean intensity 5×10^{-4} A/m) than in the *Scaglia variegata* (mean intensity 4×10^{-3} A/m), but there are large-intensity fluctuations within each formation. Magnetic susceptibility remains fairly constant around 10^{-4} (SI) in both formations and appears to be unrelated to the lithology (Fig. 2). The intensity of anhysteretic remanent magnetization (ARM) in the *Scaglia cinerea* matches the changes of NRM intensity. Preliminary results from rock magnetic studies still in progress indicate that the intensity variations of NRM and ARM in the *Scaglia cinerea* contain cyclically varying components. Their periods, estimated by converting stratigraphic depth (normal to bedding) to age using an average sedimentation rate of 11 m/Ma, are close to those of some astronomical cycles.

Progressive alternating field (AF) demagnetization was able to isolate at least two magnetization components in samples of each formation. They consisted of a low-coercivity component below about 10–20 mT and a high-coercivity component with coercivities up to 50–70 mT (Fig. 4a,d). The AF demagnetization vector diagrams do not lead to the origin, indicating the presence of a higher coercivity component that could not be eliminated successfully due to the acquisition of spurious components. Stepwise thermal demagnetization (Fig. 4b,c,e) defined a component with unblocking temperatures below 150–250°C and a component with higher unblocking temperatures up to at least 580°C. Magnetization components with both normal polarity (Fig. 4a–c) and reversed polarity (Fig. 4d,e) could be defined by either method. AF demagnetization was used preferentially for *Scaglia cinerea* samples, which showed a tendency to acquire viscous components after heating to high temperatures; thermal demagnetization was used for *Scaglia variegata* samples. The component with high unblocking temperatures or high coercivities was interpreted to be the characteristic remanent magnetization (ChRM) in each sample.

4. Paleomagnetic directions

The ChRM directions clearly display opposite polarities but the scatter is large in both polarity groups (Fig. 5a). In particular, many samples have declinations or inclinations that are intermediate between normal and reverse directions, so that a preliminary polarity interpretation contains many mag-

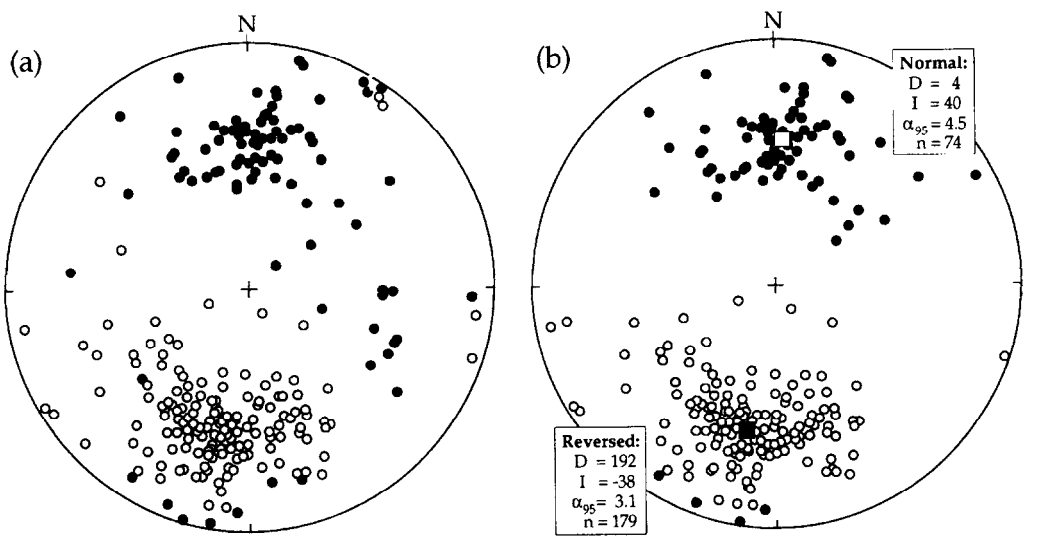


Fig. 5. Equal-area stereograms of ChRM directions after simple bedding tilt correction: (a) showing groups of normal and reversed polarities and scattered directions due to misoriented samples; (b) after correcting misoriented samples. Dots and circles represent projections on the lower and upper hemispheres, respectively.

netozones of uncertain polarity. The scattered directions are mainly due to sample misorientation. This was evident in some samples from the direction of the 'soft' component. Normally close to the present field direction, in misoriented samples it had a different direction.

The misorientation of a few specimens occurred while sub-sampling the core. Other sample misorientations were apparently due to core handling after coring. When removed from the core liner, the core sections were laid in 1 m long semi-cylindrical plastic trays, maintaining alignment of the bedding laminations visible on the core surface. Many sections consisted of smaller segments, some of which were evidently misoriented at this stage. An apparently correct alignment is possible for a core segment that has been inverted accidentally about the horizontal strike of the bedding. In order to confirm the identification of misoriented samples we applied cluster analysis to the entire set of ChRM directions.

Four clusters were assumed, corresponding to correctly oriented normal and reverse polarities and misoriented normal and reverse polarities, respectively. Each direction is associated with a cluster in such a way as to minimize the deviations within each cluster and to maximize the distances between the cluster means. The four optimized data clusters consist of two nearly antipodal groups of normal and reversed samples, and two groups with directions that can be explained by the accidental inversion of core segments. An orientation correction was applied to these samples, provided no correctly oriented samples came from the same segment; otherwise the anomalous direction was retained.

Table 1
Vector mean ChRM directions and statistical parameters [29] for normal and reversed polarity MASSICORE samples

Sample set	N	D_m	I_m	α_{95}	R	k
Normal	74	3.6	40.1	4.5	68.93	14.41
Reversed	179	191.9	−38.4	3.1	164.50	12.28
All	253	9.5	39.0	2.5	233.10	12.67

The vector means and statistical parameters of the resulting set of ChRM directions, corrected for bedding tilt by simple rotation about a horizontal axis, are given in Table 1. The ChRM directions form two nearly antipodal groups (Fig. 5b). The circles of confidence about the mean of the normal-polarity directions and the antipode of the reversed-polarity directions overlap partially. The angle between the normal-polarity mean and the antipode of the reversed-polarity mean is 6.5°, but each mean direction lies outside the confidence circle of the other. An F-ratio test [24] confirms that the mean directions are statistically different at the 95% confidence level (computed F-ratio = 4.2, null hypothesis $F_o = 3.0$), although the difference is marginal. The vector mean direction of the 253 usable samples from the MASSICORE is $D = 10°$, $I = 39°$ ($\alpha_{95} = 2.5°$). A mean direction with $D = 13°$, $I = 45°$ ($\alpha_{95} = 7.4°$) was previously reported for 29 oriented samples from the Massignano quarry [13]. Again, the confidence circles of the Massignano quarry and MASSICORE mean directions overlap; the MASSICORE mean lies inside the confidence circle of the quarry mean, but the

Table 2
Analysis of directions and poles for MASSICORE samples (site coordinates: latitude 42.54°N, longitude 13.59°E)

Analysis of directions										
	N	D_m	I_m	α_{95}	R	k	λ_p (°N)	ϕ_p (°E)	$\Delta\lambda$	$\Delta\phi$
Oligocene	114	8.2	41.1	3.7	105.00	12.56	69.8	171.4	2.7	4.5
Eocene	139	10.5	37.2	3.3	128.26	12.85	66.5	168.2	2.3	3.9
All samples	253	9.5	39.0	2.5	233.10	12.67	68.0	169.5	1.8	2.9
Analysis of poles										
	N	R	k	λ_p (°N)	ϕ_p (°E)	α_{95}				
Oligocene	114	103.84	11.13	71.0	170.4	3.9				
Eocene	139	128.31	12.91	67.2	168.0	3.3				
All samples	253	232.02	12.01	68.9	169.0	2.5				

converse is not the case. The F-ratio test shows that the 6.5° separation of the quarry and MASSICORE mean directions is not significant ($F = 1.96$; $F_o = 3.0$).

The pole positions calculated from MASSICORE samples are given in Table 2. The area of investigation belongs to the Adriatic promontory of the African plate [25,26], for which there is a notable absence of paleomagnetic data of Oligocene and Eocene age. Besse and Courtillot [27] computed an African pole position for the Eocene (age 50 Ma) by combining African paleomagnetic data with inclination data from DSDP cores and tectonically rotated terranes; it lies at 73°N, 194°E ($\alpha_{95} = 5°$). By utilizing Euler poles of relative rotation to transfer paleomagnetic data from other continents, they generated a synthetic apparent polar wander path (APWP) for each plate in their study. The 'African' synthetic pole for the Eocene (age 50 Ma) lies at 77°N, 191°E ($\alpha_{95} = 4.3°$), close to the alternative estimate. The Late Eocene pole computed from samples of both the *Scaglia variegata* and *Scaglia cinerea* in the MASSICORE is located at 67°N, 168°E ($\alpha_{95} = 3.3°$). The synthetic pole of Besse and Courtillot for the Early Oligocene (age 30 Ma) lies at 82°N, 155°E ($\alpha_{95} = 2.7°$); the comparable pole computed from *Scaglia cinerea* samples from the MASSICORE lies at 71°N, 170°E ($\alpha_{95} = 3.9°$).

Possible mechanisms that may contribute to the large discrepancies between the MASSICORE poles and the reference APWP are tectonic disturbance, mislocation of the reference pole, and inclination flattening. The Early Oligocene direction for the MASSICORE and the Massignano quarry section predicted from the Besse and Courtillot 'African' APWP is $D = 6°$, $I = 56°$. The observed declinations are close to the expected declination, suggesting that local tectonic rotation may be small. This contrasts with an anticlockwise rotation of 20° of the same formations in Umbria. The observed inclinations in the MASSICORE

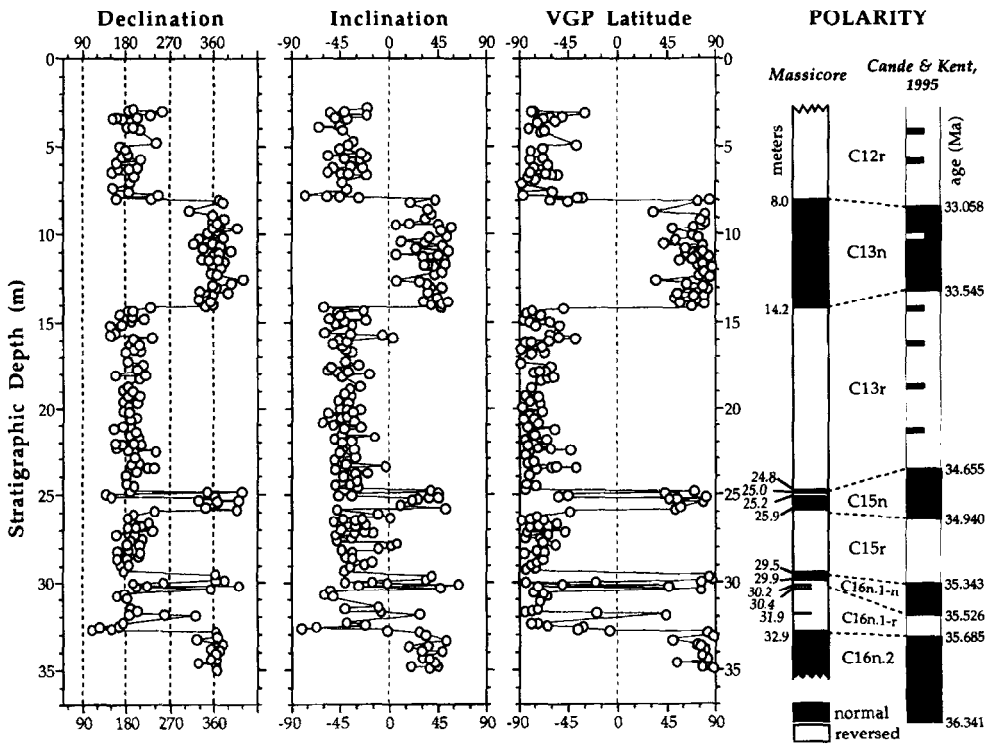


Fig. 6. Stratigraphic variations of declination, inclination and VGP latitude, and the interpreted polarity column in the MASSICORE; short horizontal bars indicate magnetozones defined by single samples. In the magnetic polarity timescale of Cande and Kent [2] short bars mark the positions of 'cryptochrons'.

are about 15° shallower than expected (in Umbria they were about 5–10° too flat). Considering that the reference directions are based on poles that are computed indirectly for the region, a possible explanation for the differences in declination and inclination is that the synthetic reference poles for the Early Oligocene and Eocene are wrongly located. However, to produce the observed inclinations the pole would have to be shifted by about 12° further from the equator; an error in the reference pole location of this magnitude seems unlikely. An alternative explanation, that the *Scaglia cinerea* and upper *Scaglia variegata* formations are affected by inclination-flattening due to post-depositional sedimentary compaction, seems more plausible. Although the flattening spoils the validity of the MASSICORE paleomagnetic directions for tectonic purposes, it does not alter their usefulness for magnetostratigraphy and related applications.

The declinations after inverting reversed samples show no significant variation with depth in the core. Likewise, inclinations remain constant at 38–40° in most of the core. However, in the depth range 25–35 m, corresponding to the top of the *Scaglia variegata* formation, inclinations average only about 30°. Below this zone the inclinations are again about 38–40°. The variations in inclination are a further indication of the possible influence of lithology; however, we were unable to establish their exact cause.

5. Magnetostratigraphy

The latitude of the virtual geomagnetic pole (VGP) was calculated for each sample by rotating the declinations to a north–south axis to compensate for possible tectonic rotation and by using a site latitude at the time of deposition calculated from the inclination of the core mean direction. The stratigraphic depth of each sample normal to bedding was calculated by adjusting the sample depth in the core for the bedding dip of 27°. The magnetic polarity stratig-

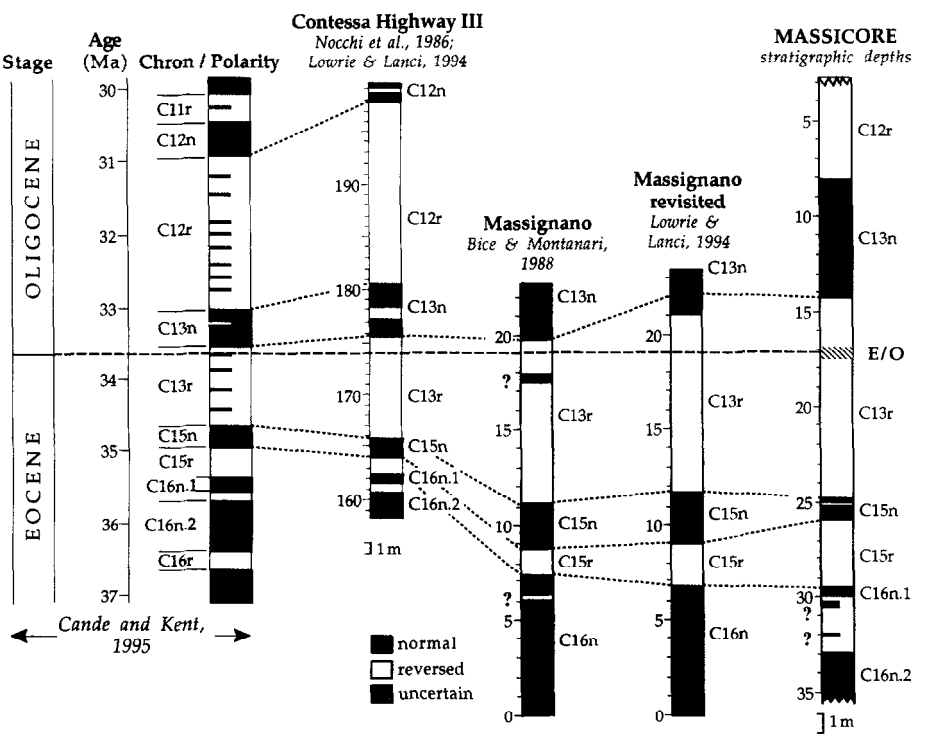


Fig. 7. Comparison of the magnetic polarity timescale of Cande and Kent [2] with the magnetostratigraphies in the Contessa and Massignano sections and the MASSICORE. Short bars indicate the positions of single-sample magnetozones in the magnetostratigraphic sections and 'cryptochrons' in the timescale.

raphy consists of ten magnetozones and three samples with polarity opposite to that of their neighbors (Fig. 6). The VGP latitude within each magnetozone generally remains in high polar latitudes. Occasional deviations into intermediate latitudes may arise from lithologic effects, unremoved secondary magnetizations, or geomagnetic field behavior.

The MASSICORE polarity sequence agrees well with the magnetic polarity timescale around the E/O boundary. Chrons C12r through C15r are clearly identifiable. Correlation of magnetozone thicknesses with corresponding chron lengths in the polarity timescale gives estimated average sedimentation rates of 11.3 m/Ma in the *Scaglia cinerea* and 8.2 m/Ma in the *Scaglia variegata*.

A thin reversed magnetozone within chron C15n at a stratigraphic depth of 25.1–25.3 m is defined by two samples, which have VGP latitudes of only 50°. If this magnetozone represents a short chron, it has no equivalent in the oceanic anomaly record, which does not indicate a 'cryptochron' at this point. The chron sequence C16n.1-n, C16n.1-r and C16n.2 contains several intermediate directions. At around 30.2 m three adjacent samples have alternating polarities and at 31.9 m there is a single normal sample. The single sample 'magnetozones' are indicated by short bars in the polarity column.

6. Conclusions

The magnetostratigraphy in the MASSICORE correlates quite well with the Massignano quarry and Contessa valley results, and with the CK95 geomagnetic polarity timescale (Fig. 7). The E/O boundary is located within chron C13r, as found in previous magnetostratigraphic investigations. However, these studies have given slightly variable estimates of the boundary location relative to the bounds of C13r. Expressing the distance of the boundary from the young end of the chron as a fraction of the chron length, the locations vary from C13r(.17) to C13r(.5). Cande and Kent [1,2] used a location at C13r(.14) as calibration level for their timescales.

The location of the E/O boundary in different sections reflects small variations in the fossil assemblages. The accepted definition is by the last occurrence (LO) of the planktonic foraminifera *Hantkeninidae* [28]. The poor state of preservation in the Contessa valley sections posed severe problems for identifying biostratigraphic events. In the Contessa quarry section [9] the E/O boundary was placed at C13r(.5) on the basis of the LO of *Globorotalia cerroazulensis*, but this location was marred by the presence of a fault near the boundary which cut out the latest Eocene fossil zones. In the nearby, more complete Contessa Highway III section [10] the boundary was located at C13r(.29). At DSDP site 522 the LO of *G. cerroazulensis* located the boundary at approximately C13r(.15) with respect to the magnetostratigraphy [6]. The LO of *Hantkeninidae* in the Massignano GSSP placed the E/O boundary at C13r(.17) relative to the magnetostratigraphy of Bice and Montanari [11]. Lowrie and Lanci [13] found additional reversed samples above the boundary in this section, but a large gap to a single normal sample at the top of the section prevented exact location of the boundary within the polarity chron; it is estimated to lie in the range C13r(.27)–C13r(.36). Paleontological analyses have not yet been made on the MASSICORE. However, the excellent correlation of the core with the Massignano quarry section using biotite-rich layers (Fig. 3) indicates that the E/O boundary in the MASSICORE should be located close to C13r(.28).

There is a discrepancy between the definitions of C16n.1–C16n.2 in the Massignano quarry and MASSICORE sections. This may be due to wider spacing of quarry samples and poorer quality of their magnetizations. Differences between relative thicknesses of the magnetozones in the core and corresponding polarity chrons in the timescale may be due to variations in sedimentation rate, as suggested by lithologically related variations of remanent magnetization intensities (Fig. 2). A possible short chron within C13n, where a 'cryptochron' is suggested [1], is found in the Contessa section but not in the others. A short chron interpreted near the E/O boundary within C13r [11] and close to an indicated 'cryptochron' was not confirmed by intensive reexamination of the Massignano quarry section [13] or by the magnetostratigraphy of the MASSICORE.

Some of the magnetostratigraphic sections in Fig. 7 contain single samples with polarity opposite to that of neighboring samples. We are unable to establish if the odd directions are due to local disturbance,

misorientation, remagnetization, or if they represent 'cryptochrons'. However, none has been confirmed in an independent overlapping section. On the principle that an event should be regarded as dubious unless its presence is confirmed in contemporaneous sections, the 'single-sample' magnetozones cannot be accepted unequivocally as polarity chrons.

On the other hand, the magnetostratigraphic sections do not contain magnetozones where 'cryptochrons' are suggested by the marine magnetic record, while single-sample magnetozones in the sections have no counterparts in the timescale. Consequently, the magnetostratigraphic data favor the interpretation that 'tiny wiggles' in the marine magnetic anomaly record are not due to short polarity chrons but probably represent geomagnetic paleointensity fluctuations [15].

Acknowledgements

We thank F. Heller, J. Ogg and an anonymous reviewer for constructive criticisms of the manuscript. Partial support for L. Lanci from a stipend of the University of Trieste and for A. Montanari from ARMINES (Paris) is gratefully acknowledged. Publication number 920, Institut für Geophysik, ETH Zürich. *[RV]*

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